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SECTION A: Multiple Choice Questions

1. Which of the following sorting algorithms has the best worst-case time complexity?
(A) Quick Sort
(B) Merge Sort
(C) Bubble Sort
(D) Insertion Sort
2. In Python, which built-in data type is immutable?
(A) list
(B) dict
(C) tuple
(D) set
3. Consider a binary search tree with n nodes. The maximum height of such a tree in the worst case scenario occurs when the elements are inserted in sorted order, resulting in:

a degenerate or pathological tree structure having a depth of:

- (A) $O(\log n)$
- (B) $O(n)$
- (C) $O(n \log n)$
- (D) $O(1)$

4. State whether True or False:

HTTP is a stateless protocol by design.

ANSWER KEY & SOLUTIONS

- 1. (B)
- 2. (C)
- 3. (B)
- 4. True